

Manifold-Informed Architecture Benchmark

CIFAR100: 100 classes, 3,072D input

Generated: 2026-08-04 03:33:58 | Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD

waverider @ 6339d08 (-abbrev-re 6339d08512638a9eee199cce4244b7362b15e2a4) | 2026-08-04 03:31:33
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Best: ManifoldResNet-UB ($w^*=118$) test acc = 0.3829 ± 0.0377
vs Standard: +0.3308 (33.08 pp) $5.8\times$ fewer params $42.5\times$ acc/Kparam
Manifold: 3,072D $\rightarrow$ 19D (99.4% noise dimensions)

Experimental Setup

Parameter	Value
Dataset	CIFAR100
Input dimensionality	3,072D
Classes	100
Intrinsic dim (d)	19
Variance threshold (τ)	0.9
Epochs	30
Trials	3

Manifold Discovery

Local PCA over the training set, $k=?$ neighbors. Intrinsic dimensionality d is taken as the maximum per-class maximum at $\tau = 0.9$, clamped to $n_{\text{classes}} = 100$.

τ	Mean d	Std	Min	Max	Noise (%)
0.95	18.9	0.9	15	21	99.4
0.90	15.7	1.1	11	18	99.5
0.85	13.3	1.1	9	16	99.6
0.80	11.5	1.1	8	14	99.6

Per-Class Intrinsic Dimensionality

Showing the 10 highest- and 10 lowest-dimensional classes of 100 total, sorted by mean d .

Class	Mean d	Std	Min	Max
51	18.0	0.0	18	18
81	18.0	0.6	17	19
48	17.8	0.4	17	18
13	17.6	0.5	17	18
14	17.6	0.5	17	18
66	17.6	1.0	16	19
6	17.4	0.5	17	18
37	17.4	0.5	17	18
58	17.4	0.5	17	18
43	17.2	0.4	17	18
...				
94	14.2	1.0	13	15
61	14.0	0.9	13	15
9	13.8	0.4	13	14
24	13.8	0.7	13	15
73	13.4	0.8	13	15
67	13.0	0.9	12	14
69	13.0	0.9	12	14
23	12.4	0.5	12	13
60	12.4	0.5	12	13
71	11.0	0.9	10	12

Architecture Comparison

Architecture	Params	Test Acc ($\mu \pm \sigma$)	Loss	Acc/Kparam
Standard (1024→512)	3,722,852	0.0521 ± 0.0048	33499.9707	0.0000
Manifold (2d→d, d=19)	317,400	0.0833 ± 0.0043	12.7129	0.0003
Manifold + ManifoldAdam (d=19)	317,400	0.0614 ± 0.0039	11.9235	0.0002
ResNet (Adam)	50,948	0.3755 ± 0.0089	2.4784	0.0074
ManifoldResNet-d (d=19)	19,176	0.3116 ± 0.0068	2.6885	0.0162
ManifoldResNet-d+C (d=19)	29,276	0.3051 ± 0.0058	2.8108	0.0104
PCA→d*→C→C (d=19)	12,100	0.0966 ± 0.0048	4.3031	0.0080
ManifoldResNet-2d (2d=38)	70,742	0.3719 ± 0.0195	2.5369	0.0053
PCA(100) Whitney(2d=38)→100	7,738	0.1502 ± 0.0019	3.8427	0.0194
Intrinsic Dim (PCA→19D→output)	2,380	0.1212 ± 0.0052	3.9027	0.0509
	644,262	0.3829 ± 0.0377	4.1482	0.0006
ManifoldResNet-UB (w*=118) ★				
UB-PCA-MLP (→119→PCA→119→100)	391,967	0.1309 ± 0.0021	4.0014	0.0003

★ = best performing architecture

Key Findings

- **Best architecture:** ManifoldResNet-UB ($w^*=118$) — test accuracy 0.3829 ± 0.0377
- **vs Standard:** +0.3308 (33.08 pp) accuracy gain
- **Parameter reduction:** $5.8\times$ fewer parameters (644,262 vs 3,722,852)
- **Parameter efficiency:** $42.5\times$ improvement (0.0006 vs 0.0000 acc/Kparam)
- **Manifold compression:** 3,072D $\rightarrow$ 19D (99.4% of ambient dimensions carry no structure)

Result Figure

Provenance

Item	Value
Repository	Flux-Frontiers/waverider
Commit	6339d08 (-abbrev-re 6339d08512638a9eee199cce4244b7362b15e2a4)
Commit date	2026-08-04 03:31:33 +0000
Commit message	docs(benchmarks): build and track the 11 typeset report PDFs
Python	3.12.3
TensorFlow	2.16.2
Device	Metal GPU (/physical_device:GPU:0)
Host	vm
OS	Linux-6.18.5-fc-v18-x86_64-with-glibc2.39
Machine	Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD
Generated	2026-08-04 03:33:58

CIFAR-100: Manifold-Informed Architecture ($d=100$) vs Standard
3,072D color images → manifold discovery → architecture | 100 fine-grained classes | run time: 381s | 2026-05-24 13:54

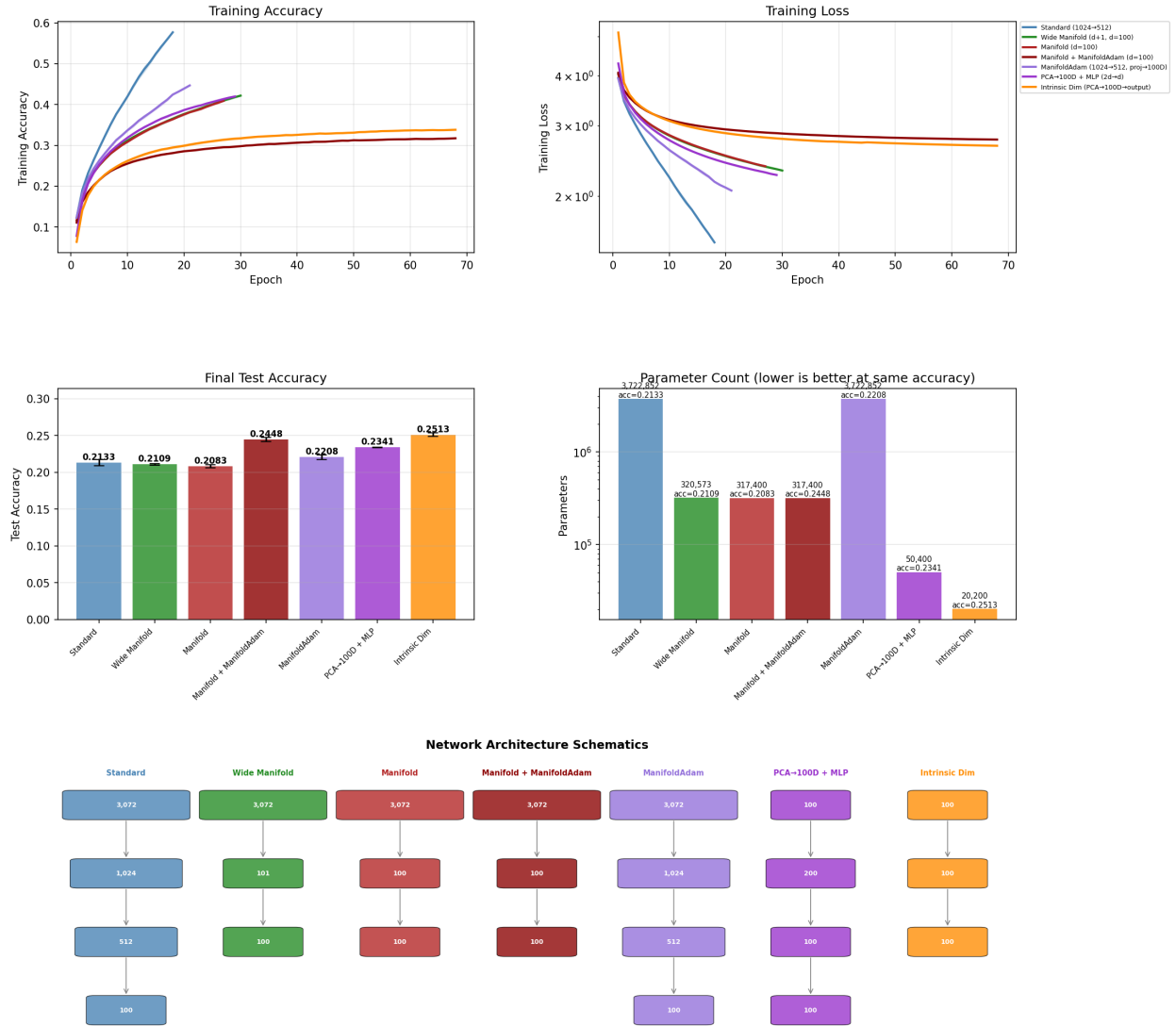


Figure 1: CIFAR100 manifold-informed vs. standard architecture comparison. $d = 19$, 30 epochs, 3 trials.